Subject: Final Project Proposal – Numerov Method for Quantum Bound States

I would like to propose a final project focused on solving the one-dimensional time-independent Schrödinger equation using the Numerov method.

The goal of the project is to implement a numerical solver for bound states in various potentials by combining the Numerov integration scheme with a shooting method. The project will focus on symmetric potentials, allowing the use of parity (even/odd states) to simplify the boundary conditions.

The main objectives are:

- Implement the Numerov method for second-order differential equations.
- Develop a shooting method with bisection to determine eigenvalues.
- Solve for bound states in several potentials:
  - Infinite square well (validation against exact solutions)
  - Harmonic oscillator (validation against exact solutions)
  - Quartic double-well potential (tunneling and energy splitting)
  - Finite square well (additional nontrivial test case)
- Normalize and analyze the resulting wavefunctions.
- Perform convergence studies with respect to grid spacing and domain size.
- Analyze physical behavior such as parity, node structure, and tunneling-induced energy splitting.

The project will include:

- A modular Python implementation (Numerov solver, shooting method, potentials).
- Visualization of wavefunctions and probability densities.
- Convergence plots demonstrating numerical accuracy.
- Parameter studies (e.g., double-well barrier dependence).
- Comparison with analytical solutions where available.
- A short report summarizing methods, results, and numerical accuracy.

This project connects directly to numerical ODE methods and eigenvalue problems discussed in the course, particularly the use of shooting methods for boundary value problems.

Best regards,
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